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UTS Elektronika Komunikasi

1. Rancang pendla LC untuk menangkap stasiun radio yang memancar dengan frekuensi 1602 kHz gunakan kapasitor 1nF

Dik : $f = 1602\ kHz$
$C = 1nF$
Dit : Rancang pendla LC
Jawab

$$\omega = \frac{1}{\sqrt{LC}}$$

$$2\pi f = \frac{1}{\sqrt{LC}}$$

$$2\pi\ 1602 \times 10^3 = \frac{1}{\sqrt{L.1 \times 10^{-9}}}$$

$$10.06 \times 10^6 = \frac{1}{\sqrt{L.1 \times 10^{-9}}}$$

$$(10.06 \times 10^6)^2 = \frac{1}{L.1 \times 10^{-9}}$$

$$101 \times 10^{12} = \frac{1}{L.1 \times 10^{-9}}$$

$$\frac{1}{L} = 101 \times 10^{12} . 1 \times 10^{-9}$$

$$L = \frac{1}{101 \times 10^3}$$

$$L = 9.90 \times 10^{-6} = 9.9\ \mu H$$

2 Rancanglah sebuah penyesuai impedansi tipe L untuk menyesuaikan tahanan sumber 10 Ω dan tahanan beban 100 Ω pada frekuensi 50 MHz. Asumsikan bahwa sinyal DC tidak ikut ditransfer ke beban

Dik : $R_s = 10\ \Omega$
$R_f = 100\ \Omega$
$f = 50\ MHz$
Dit : penyesuai impedansi tipe L

Jawab

$$Q_s = Q_p = \sqrt{\frac{R_p}{R_s} - 1}$$

$$= \sqrt{\frac{100}{10} - 1}$$

$$= \sqrt{9} = 3$$

$$X_s = Q_s R_s = 3 \; 10 = 30$$

$$X_p = \frac{R_p}{Q_p} = \frac{100}{3} = 33,3$$

$$\omega L = 30$$

$$L = \frac{30}{\omega} = \frac{30}{2\pi f} = \frac{30}{2\pi \; 50 \times 10^6} = 1,059 \times 10^{-7} = 0,1059 \; \mu H$$

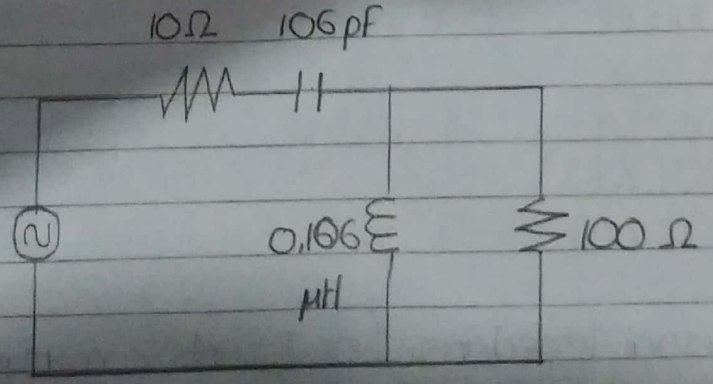
$$\frac{1}{\omega C} = 33,3$$

$$\frac{1}{C} = 33,3 \cdot 2\pi f$$

$$\frac{1}{C} = 33,3 \; 2\pi \; 50 \times 10^6$$

$$C = \frac{1}{33,3 \cdot 2\pi \cdot 50 \times 10^6}$$

$$C = 1,06 \times 10^{-10} = 106 \; pF$$



3. Rancang ulang penyesuai impedansi soal no 2 diatas dengan type T dan faktor kualitas sebesar 15.

Dik : $R_p = 100\ \Omega$ faktor kualitas = 15

$R_s = 10\ \Omega$

$f = 50\ MHz$

Dit : penyesuai impedansi type T

$$Q = \sqrt{\frac{R_p}{R_s} - 1}$$

$$R = R_{kecil}(Q^2 + 1)$$

$$= 10(226)$$

$$= 2260\ \Omega$$

$$X_{S1} = QR_s = 15 \quad 10 = 150\ \Omega$$

$$X_{p1} = \frac{R}{Q} = \frac{2260}{15} = 150.67\ \Omega$$

$$Q_2 = \sqrt{\frac{R}{R_p} - 1} = \sqrt{\frac{2260}{100} - 1} = 21,6$$

$$X_{p2} = \frac{R}{Q_2} = \frac{2260}{21,6} = 104,6\ \Omega$$

$$X_{S2} = Q_2 R_p = 21,6 \cdot 100 = 2160\ \Omega$$

4 Rancang Sebuah LPF yang memenuhi spesifikasi seperti gambar di bawah

Dimana nilai $R_S = 75\ \Omega$ dan $R_L = 150\ \Omega$

Dik : $f_C = 15\ kHz$
$R_S = 75\ \Omega$
$R_L = 150\ \Omega$

Dit : LPF

Jawab.

$\frac{R_S}{R_L} = \frac{75\ \Omega}{150\ \Omega} = 0,5$ atau $\frac{R_L}{R_S} = \frac{150\ \Omega}{75\ \Omega} = 2$

$\frac{f}{f_C} = \frac{60\ kHz}{15\ kHz} = 4$

$L_1 = \frac{R_L L_4}{2\pi f_C} = \frac{150\ 0,218}{2\pi\ 15\ kHz} = \frac{32,7}{94,2 \times 10^3} = 0,347\ \mu H$

$C_1 = \frac{C_4}{2\pi f_C R_L} = \frac{2,452}{2\pi\ 15kHz\ 150} = \frac{2,452}{14130 \times 10^3} = 17,35\ \mu H$

$L_2 = \frac{R_L L_4}{2\pi f_C} = \frac{150\ 0,883}{2\pi\ 15kHz} = \frac{132,45}{94,2 \times 10^3} = 1,40\ \mu H$

$C_1 = \frac{C_4}{2\pi f_C R_L} = \frac{3,187}{2\pi\ 15kHz\ 150} = \frac{3,187}{14130 \times 10^3} = 22\ \mu H$